

# UrbanFix: A Closed-Loop, AI-Validated Urban Governance Platform with a Public Accountability Registry and Privacy-Preserving Research Data Access

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**Abstract**—Most digital grievance systems classify complaint text and track tickets. They rarely check the photo a citizen attaches, their urgency estimates cannot be traced to inputs, and only the administration sees the records. We present UrbanFix, an urban governance platform that follows a complaint from the citizen’s photo to a verified, publicly recorded repair and on to research data. Citizens file without an account using an email code. A YOLOv8 model trained for the chosen category localises the defect in each photo, and a weighted questionnaire gives an urgency score anyone can recompute. A seven-stage workflow enforced on the server requires ward-level assignment, photo proof and a receipt before closure. Every complaint is published without personal data, and approved researchers get logged, time-limited access to an anonymised projection. Validation mAP50 is 0.724 for potholes and 0.502 for litter (masks), and 0.908 for open manholes and 0.729 for graffiti (boxes), at 19 to 33 ms per image on a laptop CPU. When the model is unsure, it flags the report for a human reviewer.

**Index Terms**—e-governance, smart cities, civic technology, object detection, YOLOv8, human-in-the-loop AI, open data, privacy-preserving data sharing

## I. INTRODUCTION

In India, pothole-related road accidents rose from 3,713 in 2020 to 5,432 in 2024, an increase of 46%, and pothole-related accidents killed 8,963 people between 2018 and 2022 [1], [2]. The national grievance portal CPGRAMS received between 18.7 and 26.2 lakh grievances a year from 2019 to 2024 [3], [4], and in FY 2023–24 State and Union Territory governments, which handle municipal complaints, took 119 days on average to dispose of one, against 16 days for central ministries [5] (Fig. 1).

Fig. 1(a) shows pothole accidents dipping slightly in 2021, peaking at 5,840 in 2023 and staying above 5,000 in 2024, so the problem grew over the five years. In Fig. 1(b) the light bars are the backlog carried over from the previous year, which fell from 10.7 lakh in 2020 to 3.1 lakh in 2024; the portal now clears most of what it receives. The dark bars are new grievances, and they reached their highest value, 26.2 lakh, in the same year. Faster disposal has not reduced the number

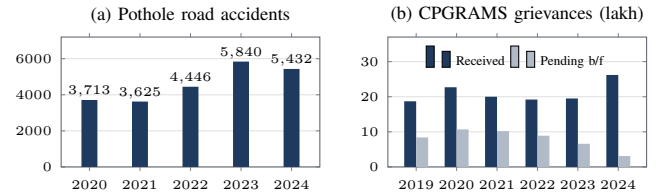


Fig. 1. Motivation: (a) pothole-related road accidents in India [2]; (b) grievances received on CPGRAMS and the backlog brought forward each year [3], [4].

of people who need to complain. We therefore look at what happens to a complaint after it is filed: whether its evidence is checked, whether the repair is proven, and whether anyone outside the administration can see the result.

Recent systems route complaint text [7], [12], predict urgency [8], classify waste images [9] or filter unsafe photos [10]. None checks whether a photo shows the reported defect, their urgency cannot be traced to an input, and their records stay with officials or the filer. UrbanFix runs each complaint through six steps in one system (Fig. 2). Our contributions are:

- 1) A YOLOv8 check [18] for each category, offered only where a public dataset with localised labels exists.
- 2) Prioritisation that puts the questionnaire score and the model’s evidence in front of a supervisor, who decides; the model cannot reject a report.
- 3) A workflow modelled as a guarded state machine with ward-level assignment, photo proof and an append-only history.
- 4) A public registry with identity removed on the server, and a research gateway with scoped, expiring and logged access to pseudonymised records.

Fig. 2 reads clockwise from the top. A citizen reports without an account, and the model for the chosen category checks the photo. A supervisor then sets the priority with the

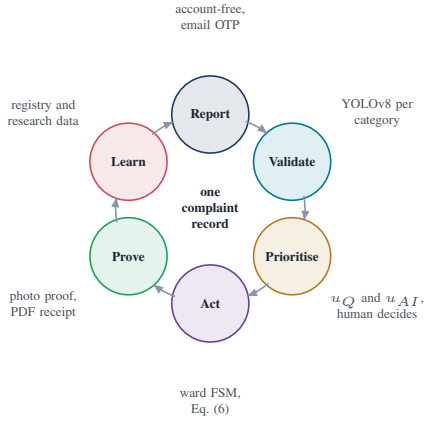


Fig. 2. The six steps every complaint passes through in UrbanFix.

TABLE I  
CAPABILITIES OF REVIEWED SYSTEMS AND URBANFIX (UF)

| Capability              | [6] | [7] | [8] | [9] | [10] | [11] | [12] | UF |
|-------------------------|-----|-----|-----|-----|------|------|------|----|
| Account-free filing     | ○   | ●   | ○   | ○   | ○    | ○    | ○    | ●  |
| Photo evidence          | ●   | ○   | ●   | ●   | ●    | ●    | ●    | ●  |
| Photo-defect validation | ○   | ○   | ●   | ○   | ○    | ○    | ○    | ●  |
| Urgency scoring         | ○   | ○   | ●   | ○   | ○    | ●    | ○    | ●  |
| Traceable urgency       | ○   | ○   | ○   | ○   | ○    | ○    | ○    | ●  |
| Automated routing       | ●   | ●   | ●   | ●   | ●    | ●    | ●    | ●  |
| Field proof of repair   | ●   | ○   | ○   | ○   | ○    | ○    | ○    | ●  |
| Resolution receipt      | ●   | ○   | ○   | ○   | ○    | ○    | ○    | ●  |
| Public registry         | ○   | ○   | ○   | ○   | ○    | ○    | ○    | ●  |
| Open research data      | ○   | ○   | ○   | ○   | ○    | ○    | ○    | ●  |

● supported    ● partial    ○ not supported or not reported

questionnaire score  $u_Q$  and the model's suggestion  $u_{AI}$  side by side. Field staff carry out the work inside the state machine of Eq. (6) and prove the repair with photos, the citizen receives a PDF receipt, and the closed record feeds the public registry and the research dataset. The centre of the ring is a single complaint record. Every step reads and appends to the same document, so the history the public and researchers see is the one staff produced while working.

## II. RELATED WORK

Infra-Alert [6] geotags photos and generates PDF reports. CitiWatch [7] routes text at 96.8% accuracy, and Cholke *et al.* [8] add multi-task urgency prediction and duplicate grouping. ReGenX [9] classifies waste photos, CivicMate [10] screens unsafe images, CivicX [11] compares its classifier with officers (91.4% vs. 94.1%), and Smart City Concern [12] uses sentiment as a proxy for urgency. Table I lists the gaps; the largest is that none checks the photo against the reported defect. Localised datasets now exist for road damage [13], [14], litter [15], open manholes [16] and graffiti [17].

## III. SYSTEM ARCHITECTURE

One Expo client serves all six roles, and the server decides which screens a user sees (Fig. 3). Each request passes an Express pipeline that re-checks the account, any pending

invitation and research expiry on every call. Intake sends photos to a separate FastAPI service only after returning the tracking ID, so a slow or failed model call cannot affect the filing. Each MongoDB complaint document embeds its status history and AI analysis.

Fig. 3 reads from top to bottom. The six actors use the same client, which gives each of them one of five role shells; citizens and the public share one. Every call then passes the seven gateway stages in order, so a request that fails authentication never reaches validation or a controller. Seven domain services sit below the gateway. The research gateway is drawn in purple because it is the only path to bulk data. In the lower left tier, a router sends each photo to one of four YOLOv8n models, and a post-processing step computes the evidence of Eqs. (3)–(5). The lower right tier holds MongoDB and the services for files, email, receipts and scheduled expiry notices. The two dashed arrows are the only asynchronous paths: photos go to the model after the citizen already has a tracking ID, and the result returns as `aiAnalysis` for triage.

## IV. METHODOLOGY

### A. Category selection by dataset availability

Let  $\mathcal{K}$  be the candidate complaint categories. A category  $k$  is offered only if a public dataset  $\mathcal{D}_k$  of street-level images with localised labels (boxes or masks), documented provenance and a usable licence exists:

$$\mathcal{C} = \{k \in \mathcal{K} : \mathcal{D}_k \text{ is public} \wedge \text{localised} \wedge \text{street-level}\}. \quad (1)$$

Four of ten candidate categories qualify (Fig. 4), and the API rejects the rest.

In Fig. 4 each bar is the size of the largest public dataset we found for that category, on a log scale, and the four circles show whether it meets each criterion. Electrical poles have the second largest dataset, 25,426 images, but its labels are only partly localised and its source is undocumented. Water leakage has localised labels, but its images are not street-level and its source is also undocumented. Graffiti has only 1,022 images and meets every criterion, so it is kept. The three smallest candidates have fewer than 1,000 images; road signs and fallen trees also lack a documented source, and the streetlight data has no localised labels. Only the four categories with every circle filled satisfy Eq. (1).

### B. Traceable questionnaire urgency

Each category has five yes/no questions  $q_i$  with weights  $w_i \in \{1, 2\}$  (2 for safety-critical). With answers  $y_i \in \{0, 1\}$ , the urgency score and band are

$$U = \frac{\sum_{i=1}^5 w_i y_i}{\sum_{i=1}^5 w_i}, \quad u_Q = \begin{cases} \text{High,} & U \geq 0.6 \\ \text{Medium,} & 0.3 \leq U < 0.6 \\ \text{Standard,} & U < 0.3. \end{cases} \quad (2)$$

Dividing by  $\sum w_i$  keeps categories with different weights comparable. The citizen sees  $u_Q$  while filing, and anyone can recompute it from the stored answers (Fig. 5).

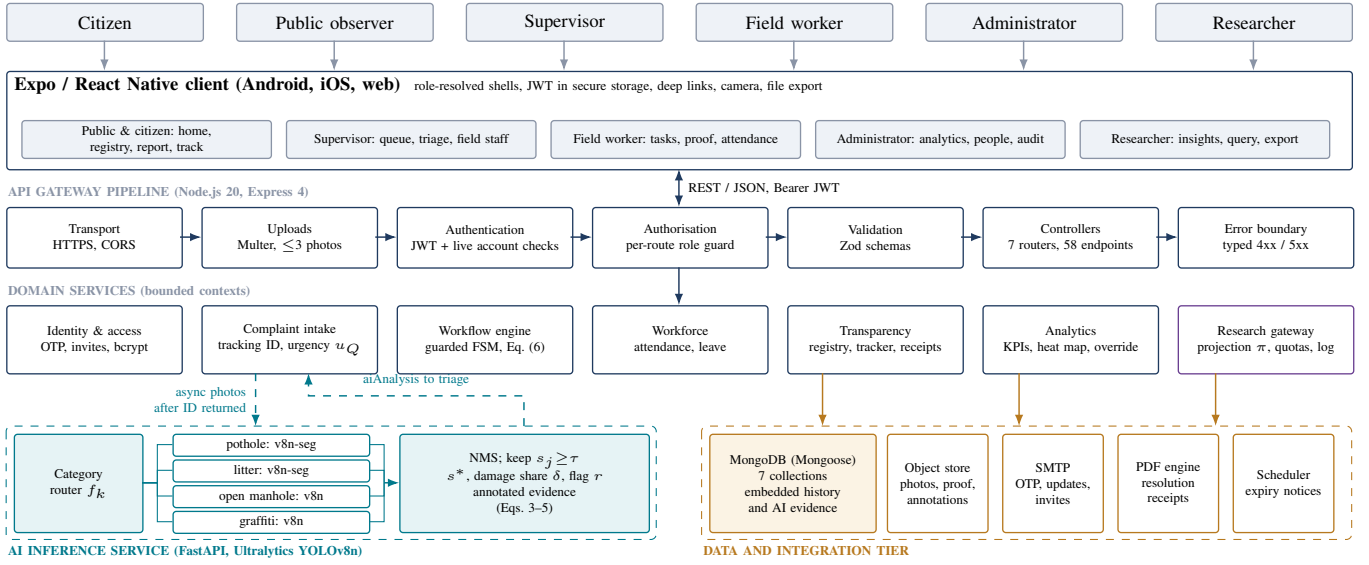


Fig. 3. Layered architecture of UrbanFix. Solid arrows are synchronous requests. Dashed arrows show the AI call, which is made asynchronously after the tracking ID has been returned. The security and privacy controls of Sections IV-D–IV-E apply to all layers.

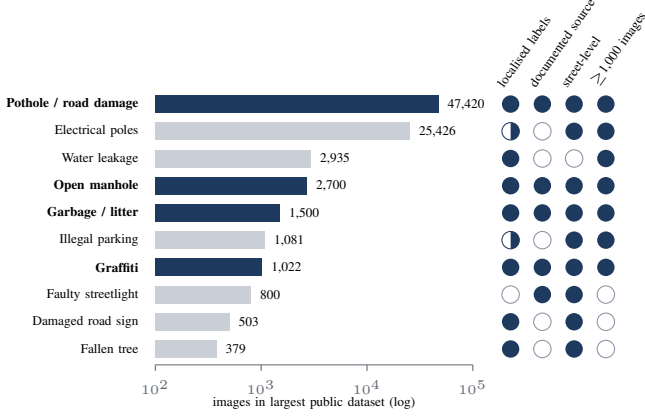


Fig. 4. Screening of ten candidate categories against Eq. (1). Dark bars mark the four categories kept; circles show each criterion met, partly met or not met.

Fig. 5 plots every score Eq. (2) can produce. The pothole and manhole questionnaires mark two questions as safety-critical, so their weights sum to 7 and  $U$  moves in steps of  $1/7$ . Litter and graffiti mark one, so the sum is 6 and the steps are  $1/6$ . Both profiles have reachable scores in all three bands, but the same answers can land in different bands. One safety-critical “yes” gives a pothole  $2/7 = 0.29$ , just short of Medium, while the same answer for litter gives  $2/6 = 0.33$  and reaches Medium. A pothole is rated High only when its “yes” answers carry at least 5 of the 7 weight points.

### C. Per-category vision validation

For a complaint of category  $k$  with photos  $x_1, \dots, x_n$  ( $n \leq 3$ ), the model  $f_k$  returns detections  $\mathcal{B}_m = f_k(x_m) = \{(b_j, s_j, M_j)\}$  with boxes  $b_j$ , confidences  $s_j$  and, for segmentation models, masks  $M_j$ . The service keeps detections

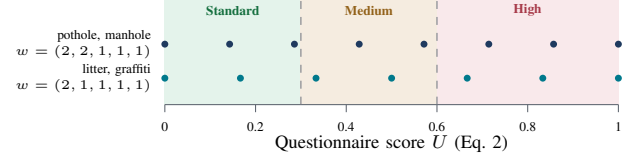


Fig. 5. All reachable questionnaire scores for the two weight profiles, with the urgency bands. For a pothole, a single safety-critical “yes” scores  $2/7 = 0.29$  and stays in Standard.

with  $s_j \geq \tau$  ( $\tau = 0.25$ ) after non-maximum suppression and computes

$$s^* = \max_m \max_j s_j, \quad (3)$$

$$\delta = \max_m \frac{|\bigcup_j M_j^{(m)}|}{H_m W_m}, \quad (4)$$

$$r = \mathbf{1}[s^* < \tau], \quad (5)$$

where  $\delta$  is the masked share of the image and  $r$  is the needsReview flag. The suggested urgency  $u_{AI}$  is  $u_Q$  raised one band when  $\delta > \theta_\delta$ . The model never changes a complaint’s stage; the supervisor decides with all of this in view (Fig. 6).

Fig. 6 shows the order of events. The questionnaire score is computed while the citizen files, and the tracking ID is returned before any inference runs, so the dashed call to the model is the only step that can be slow. The model’s detections are reduced to  $s^*$  and  $\delta$  and then tested once. If  $s^* \geq \tau$ , the service suggests  $u_{AI}$ . If not, it sets  $r = 1$  and the supervisor works from  $u_Q$  alone (the dotted line). Both branches end in the same triage box, so a photo the model could not confirm still reaches a person, with a flag, and is never rejected automatically.

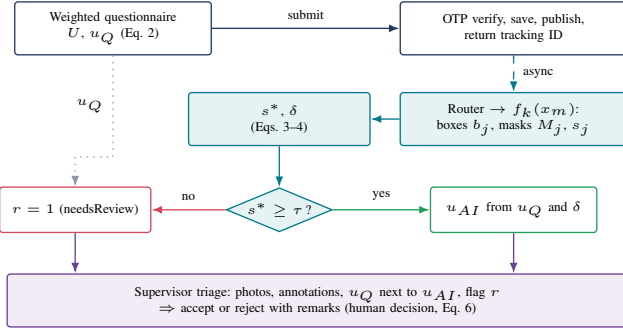


Fig. 6. AI validation and prioritisation. The supervisor sees both urgency signals; a low-confidence result sets *needsReview* and the report stays in the queue.

#### D. Accountable workflow

The workflow is a deterministic finite-state machine  $\mathcal{M} = (S, A, \Delta, s_0, F)$  (Fig. 8) with

$$S = \{\text{sub}, \text{acc}, \text{asg}, \text{wip}, \text{prf}, \text{cls}, \text{rej}\},$$

$$A = \{\text{accept}, \text{reject}, \text{assign}, \text{start}, \text{proof}, \text{close}, \text{rework}\},$$

start  $s_0 = \text{sub}$  and terminal set  $F = \{\text{cls}, \text{rej}\}$ . A transition is applied only if its guard holds:

$$\Delta(s, a) \text{ is defined } \iff s \in S_a \wedge \rho(\text{actor}) \in \mathcal{R}_a \wedge g_a(c), \quad (6)$$

where  $S_a$  and  $\mathcal{R}_a$  are the allowed source stages and roles, and the guard  $g_a$  checks remarks ( $\geq 10$  characters), the ward and leave status of the assignee, and the proof photos. Each transition appends (stage, actor, remarks,  $t$ ) to the history, and *close* emails a PDF receipt. The public sees  $\varphi(s) \in \{\text{Pending}, \text{In Progress}, \text{Resolved}, \text{Rejected}\}$ .

Fig. 8 shows every path a complaint can take. The main path runs through five transitions from submitted to closed, and Eq. (6) guards each one. Rejection is possible only from the submitted stage. After acceptance the only terminal stage left is closed, and reaching it requires proof. If the supervisor finds the proof inadequate, the dashed rework arrow returns the task to assigned, and the loop on assigned lets the supervisor hand it to another worker without changing the stage. The fill colours give the public status. A citizen sees In Progress for all four middle stages, so internal handoffs do not appear on the tracker.

#### E. Tiered disclosure and research projection

Each audience gets a fixed server-side projection of the same record (Table II). Researchers get eight fields

$$\pi(c) = (h_\kappa(c.id), k, u_Q, \varphi, \text{ward}, [t_f]_d, [t_c]_d, \Delta t), \quad (7)$$

$$h_\kappa(x) = \text{trunc}_{48}(\text{HMAC-SHA256}(\kappa, x)), \quad \Delta t = \frac{t_c - t_f}{86400 \text{ s}}, \quad (8)$$

where  $[\cdot]_d$  truncates to the day and  $\kappa$  is a server secret, so IDs join across exports but cannot be reversed. A request  $q$  under scope  $\sigma$  and expiry  $t_e \leq t_g + 180$  days is served only if

$$\begin{aligned} & t_q < t_e \wedge (\text{rows}(q) \Rightarrow \sigma = \text{records}) \\ & \wedge n_{\text{exp}}(d) \leq 5 \wedge |q| \leq 5000, \end{aligned} \quad (9)$$

TABLE II  
DISCLOSURE OF ONE COMPLAINT RECORD BY AUDIENCE

| Attribute                  | Admin | Staff | Registry | Tracker | Res.rec. | Res.agg. |
|----------------------------|-------|-------|----------|---------|----------|----------|
| Citizen identity           | ●     | ○     | ○        | ○       | ○        | ○        |
| Staff identity             | ●     | ●     | ○        | ○       | ○        | ○        |
| Record ID                  | ●     | ●     | ●        | ●       | ●        | ○        |
| Title, description, photos | ●     | ●     | ●        | ●       | ○        | ○        |
| Street location            | ●     | ●     | ●        | ●       | ●        | ○        |
| Category, urgency, status  | ●     | ●     | ●        | ●       | ●        | ●        |
| Timestamps                 | ●     | ●     | ●        | ●       | ●        | ●        |
| History with remarks       | ●     | ●     | ○        | ●       | ○        | ○        |

○ transformed: HMAC ID, ward-level, day-level or aggregated.

Fig. 9 follows a researcher from application to data. The first three steps happen once. The applicant states a purpose, an administrator sets the scope  $\sigma$  and the expiry  $t_e$ , and the invitation link is stored only as a SHA-256 hash and lapses after 72 hours. The last three steps repeat on every request. The server checks Eq. (9) and answers 403 or 429 if any part fails. A request that passes receives only the projection  $\pi(c)$ , and the request is written to the access log.

#### F. Data model

One **User** collection holds all five roles, and **Complaint** embeds its history and AI analysis (Fig. 7). Staff accounts are only ever deactivated, so history entries keep a valid actor.

Every relationship line in Fig. 7 ends at **User**. A complaint points to users in three ways: the citizen who filed it (exactly one), the worker it is assigned to (none or one), and the actor of each history event (one per event). The four collections on the right attach to **User** in the same way. A user has at most one research application and any number of access-log entries, leave requests and attendance records, and a unique index allows one attendance record per employee per day. **Otp** is joined by a dashed line because it is matched by email, not by a stored reference, and its TTL index removes each code five minutes after it is issued. Because the history and the AI evidence are embedded, each complaint loads as a single document without a join.

### V. IMPLEMENTATION AND EXPERIMENTAL SETUP

The stack is Expo 57, Node.js 20 with Express and Mongoose, and FastAPI with Ultralytics 8.4 [18]. Each model is fine-tuned from COCO YOLOv8n weights at  $640 \times 640$ , batch 16, up to 60 epochs with patience 15, on an Apple M5 laptop (Table III, Fig. 11); training took 29 to 52 minutes.

We report precision  $P = \frac{TP}{TP+FP}$ , recall  $R = \frac{TP}{TP+FN}$  and

$$\text{AP} = \int_0^1 p(r) dr, \quad (10)$$

with mAP50 at IoU 0.5 and mAP50-95 averaged over 0.50 to 0.95.

### VI. RESULTS AND DISCUSSION

#### A. Validation models

Table IV and Figs. 11–14 give the results. Open manholes score highest (box mAP50 0.908), though augmented copies

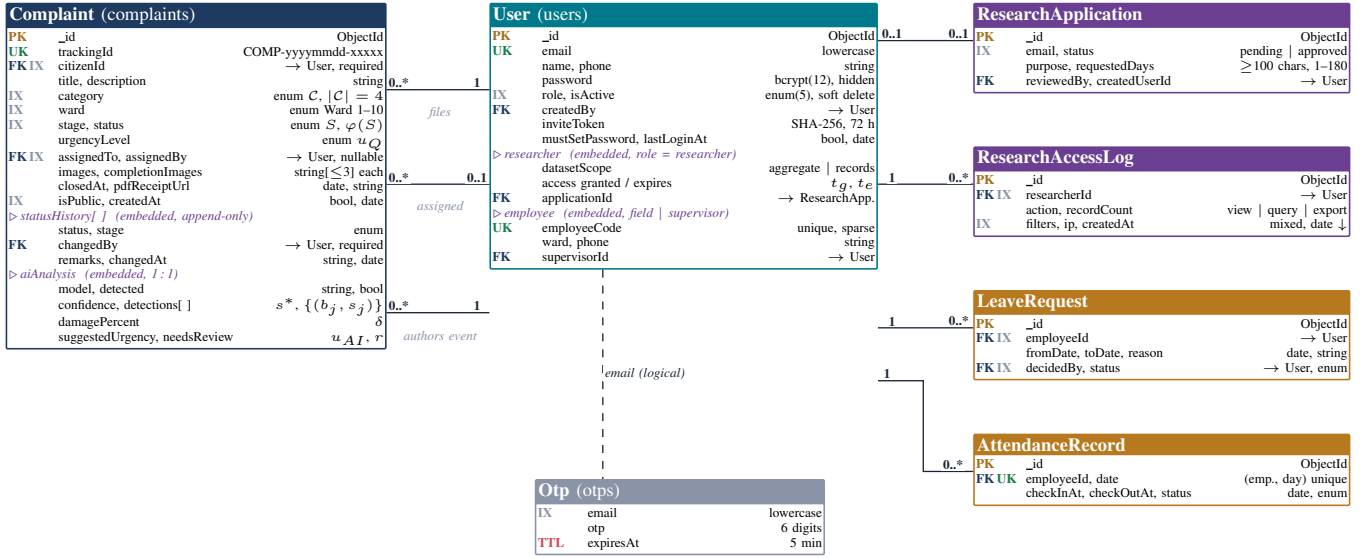


Fig. 7. Physical data model (MongoDB, Mongoose schemas). PK, FK, UK, IX and TTL mark primary keys, references, unique, secondary and expiry indexes; rows marked  $\triangleright$  begin embedded sub-documents. Multiplicities follow UML notation.

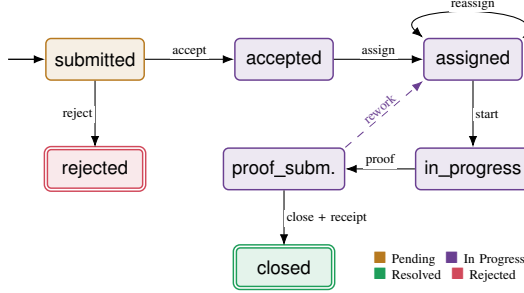


Fig. 8. Server-enforced complaint lifecycle (Eq. 6). Fill colour gives the public status  $\varphi(s)$ ; double borders mark terminal stages.

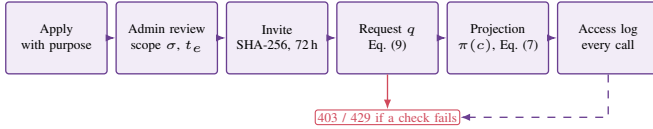


Fig. 9. Research access chain. Expiry, scope and quotas are checked on every request, and every request is logged.

TABLE III  
DATASETS AND MODELS PER CATEGORY

| Category              | Dataset           | Labels             | Train/val | Model   |
|-----------------------|-------------------|--------------------|-----------|---------|
| Pothole / road damage | Roboflow [14]     | masks              | 720/60    | v8n-seg |
| Garbage / litter      | TACO [15]         | masks <sup>†</sup> | 1214/214  | v8n-seg |
| Open manhole          | Road Hazards [16] | boxes              | 1123/480  | v8n     |
| Graffiti              | STORM [17]        | boxes              | 813/209   | v8n     |

<sup>†</sup>60 TACO classes merged into one; 1,428 of 1,500 images still available.

in that dataset may inflate it. Graffiti is precise but misses faint tags (recall 0.604), and litter is hardest (0.502) because TACO's 60 classes were merged into one. These misses are why Eq. (5) only flags reports. Inference takes 19 to 33 ms

per photo on CPU (Table IV), so no GPU server is needed. Fig. 10 shows stored outputs and app screens.

Fig. 11 shows how each model learned. The manhole model passed 0.8 by epoch 9 and stayed between 0.86 and 0.95 from epoch 13, apart from a one-epoch collapse to 0.02 at epoch 4 that it recovered from at once. The pothole curve was unstable for the first three epochs, then rose to about 0.7 by epoch 25 and stayed there. Litter and graffiti climbed slowly and steadily, to 0.50 and 0.72. Each curve ends where its run stopped, and all four had levelled off by then, so more epochs on the same data would add little; better scores will need more images or cleaner labels.

Fig. 12 counts what each model did with every labelled instance in its validation set. The manhole model found 134 instances, missed 14 and raised 29 false alarms. The pothole and graffiti models missed 27% and 30% of their instances. Litter is the outlier, with 357 instances missed against 337 found. The red segments are the cases Eq. (5) exists for: when no detection in any photo clears  $\tau$ ,  $s^* < \tau$  and the report goes to a person flagged for review.

Fig. 13 puts the four metrics side by side. Precision exceeds recall for every model, so most of their errors are misses. The gap is widest for litter (0.757 against 0.434) and graffiti (0.836 against 0.604) and almost nil for potholes (0.725 against 0.711). mAP50–95 is well below mAP50 throughout, which means the models find the defect but outline it loosely. Graffiti has the highest mAP50–95 (0.526) even though its mAP50 is lower than the manhole model's, so the graffiti boxes it does draw fit closely.

The precision–recall curves in Fig. 14 show the trade between misses and false alarms as the confidence threshold falls. The manhole curve stays above 0.89 precision until recall reaches 0.87, so that model can find most manholes with few false alarms. Pothole and graffiti keep precision above 0.85 up

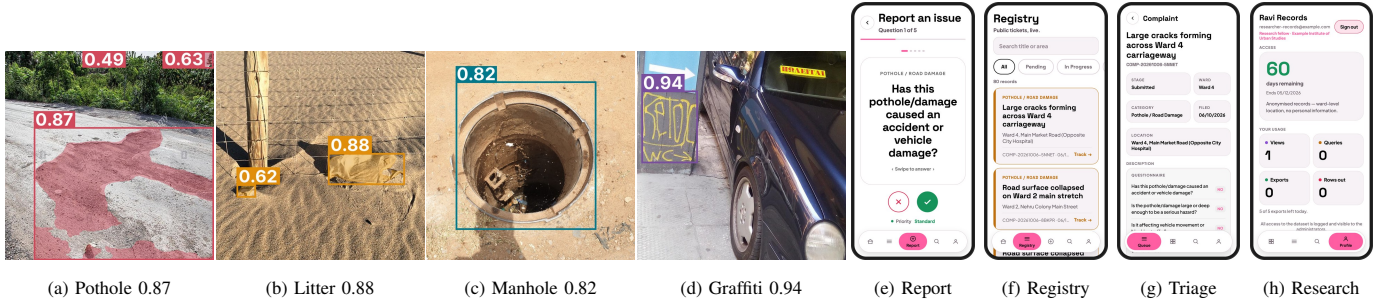


Fig. 10. (a)–(d) Stored AI evidence on validation images: predicted masks or boxes with confidence. (e)–(h) App screens: weighted swipe questionnaire with live urgency, open public registry, supervisor triage, and researcher access with expiry countdown, quota and audit notice.

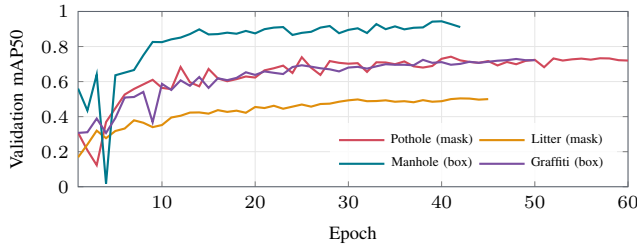


Fig. 11. Validation mAP50 per epoch. The manhole run stopped early at epoch 42.

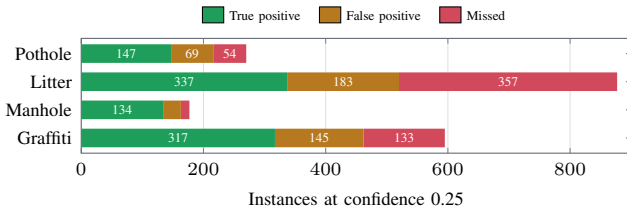


Fig. 12. Detection outcomes on the validation sets (manhole: 134, 29, 14). Missed instances (red) are what Eq. (5) routes to a person.

to a recall of about 0.55 and then decline steadily. Litter falls sharply after a recall of 0.4 and reaches zero precision at 0.72, so no threshold recovers more than about seven in ten litter instances. The area under each curve is the AP of Eq. (10), given in the legend.

Panels (a)–(d) of Fig. 10 are the evidence images the service stores with a complaint, drawn here on validation photos with the detection’s confidence beneath each. The pothole and litter models draw masks, which the damage share  $\delta$  of Eq. (4) needs; the manhole and graffiti models draw boxes. Panels (e)–(h) are four app screens: the citizen’s questionnaire with its live urgency band, the public registry, the supervisor’s triage view, and the researcher’s profile with its expiry countdown and quota.

### B. System validation and demonstration data

All sixteen API tests passed (Fig. 15). To exercise the dashboards we seeded 80 complaints over twelve months (Fig. 16): 41 were accepted, 9 rejected and 19 closed, after 8.2 to 9.2 days on average.

TABLE IV  
VALIDATION RESULTS (BEST CHECKPOINT)

| Model (metric set) | $P$   | $R$   | mAP50        | mAP50–95     | CPU ms |
|--------------------|-------|-------|--------------|--------------|--------|
| Pothole (mask)     | 0.725 | 0.711 | 0.724        | 0.426        | 33     |
| Pothole (box)      | 0.733 | 0.682 | 0.719        | 0.447        |        |
| Litter (mask)      | 0.757 | 0.434 | 0.502        | 0.317        | 24     |
| Litter (box)       | 0.748 | 0.446 | 0.521        | 0.375        |        |
| Open manhole (box) | 0.890 | 0.874 | <b>0.908</b> | 0.469        | 19     |
| Graffiti (box)     | 0.836 | 0.604 | 0.729        | <b>0.526</b> | 20     |

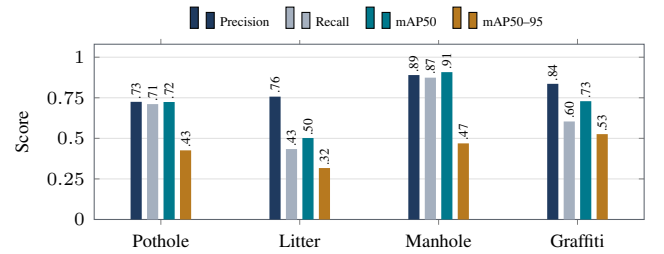


Fig. 13. Validation metrics per model (mask metrics for the segmentation models, box metrics for the detection models).

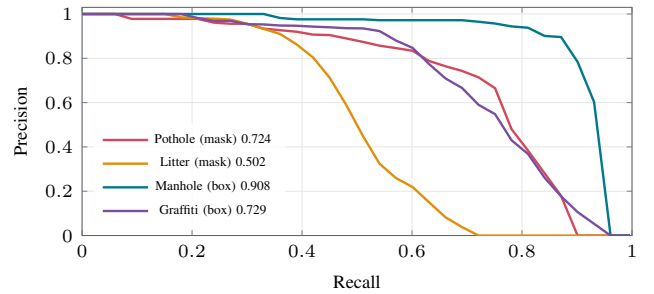


Fig. 14. Precision–recall curves on the validation sets; legend gives AP at IoU 0.5.

Fig. 15 groups the tests by the rule they check and gives the expected response in each tile. Eleven of the sixteen expect a refusal. Four cover intake: a missing email, a category outside  $\mathcal{C}$ , and a wrong or expired OTP. Five cover workflow guards: a field worker trying to accept (403), remarks shorter than ten characters, a rejection after acceptance, a cross-ward assignment, and a receipt requested before closure. The other

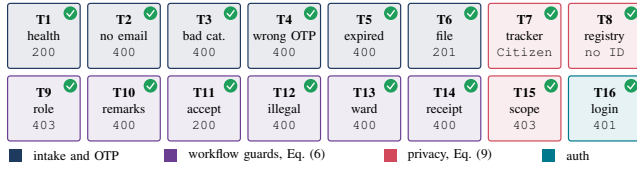


Fig. 15. Sixteen API tests against a live server, grouped by the rule they exercise; expected and observed responses matched in all cases.

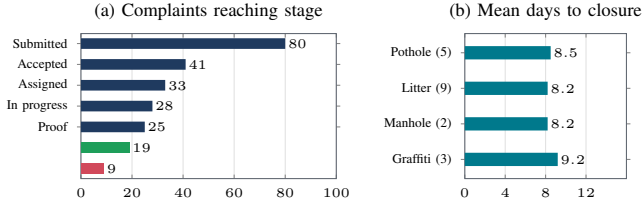


Fig. 16. Workflow on the seeded demonstration dataset ( $n = 80$ ); closed complaints per category in brackets. They exercise the dashboards only; no real ward was measured.

two are an aggregate-only researcher asking for records (403) and a failed login (401). Each refusal comes from the server, so these rules hold whatever the client sends. Of the five tests that expect success, two check privacy: they confirm that the registry response carries no citizen identity and that the tracker shows a citizen in the history only as “Citizen”.

Fig. 16(a) shows how far the 80 seeded complaints progressed. Of those, 41 were accepted and 9 rejected, leaving 30 awaiting a supervisor’s decision. Of the accepted complaints, 33 were assigned and 19 were closed with proof. Each stage holds fewer complaints than the one before because the workflow never skips a stage. Panel (b) shows mean days to closure between 8.2 and 9.2 for every category. With only two to nine closed complaints per category the differences are too small to interpret, and since the data is seeded, the panel shows only that the dashboards can compute the measure.

### C. Limitations

Common complaints such as water leakage are unsupported for lack of data, and the models have not been tested on Indian street photos. We set the questionnaire weights and  $\theta_\delta$  by hand. There is no GPS or duplicate detection, and the workflow numbers come from seeded data.

## VII. CONCLUSION AND FUTURE WORK

UrbanFix takes a civic complaint from the citizen’s photo to a recorded repair using open-source software and public data. A per-category model checks the photo, a supervisor sets the priority, and the server enforces each step before the record is published without personal data. Next we will fine-tune on local images, fit the weights to closed complaints, add GPS and duplicate detection, and run a ward-level pilot with a municipal corporation.

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